Gyutae Oh

CURRICULUM VITAE

Media System Lab (MSL), Sungkyunkwan University
Electrical and Computer Engineering
alswo740012@g.skku.edu
Portfolio
Google Scholar
Git
LinkedIn

AI graduate researcher focused on **efficient learning**, **continual learning**, **anomaly detection**, and **medical image analysis**. Currently pursuing a Ph.D. in Electrical and Computer Engineering at Sungkyunkwan University under the supervision of Prof. Jitae Shin.

EDUCATION

Sungkyunkwan University, Ph.D. in Electrical and Computer Engineering — Feb 2026 – Present
GPA: 4.5/4.5

Sungkyunkwan University, M.S. in Electrical and Computer Engineering — Feb 2024 – Feb 2026
GPA: 4.07/4.5

Korea National University of Transportation, B.S. in Computer Engineering — Mar 2019 – Feb 2024
GPA: 3.79/4.5

RESEARCH EXPERIENCE

Media System Lab (MSL), Sungkyunkwan University — Jan 2024 – Present
Advisor: Jitae Shin

Applied Machine Intelligence Lab (AMI-Lab), KNUT — Mar 2022 – Jun 2023
Advisor: Jeonghwan Gwak

MILITARY SERVICE

Republic of Korea Army, Sergeant, Honorably Discharged — 2020 – 2022
TOD operator at Gachilbong GOP, 21st Infantry Division, under a direct unit of III Corps

SELECTED PUBLICATIONS

* First author

UniPrompt-CL: Efficient Single-Inference Continual Learning for Medical Image Classification
Gyutae Oh*, Jitae Shin
WACV, 2027

RESEARCH INTERESTS

Continual Learning · Efficient Learning · Anomaly Detection · Medical Image Analysis · PyTorch · Evaluation

RESEARCH OUTPUTS

17 total papers
5 preprints · 12 peer-reviewed publications

1 patent
7 scholarships · 4 awards · 1 certification · 1 service

HONORS & AWARDS

Hanium: Companion Dog Eye Disease Detection Assistant App — Selected Works Prize (2023.11)
Hanium ICT Mentoring Program, hosted by the Ministry of Science and ICT (MSIT), Korea

Companion Dog Eye Disease Detection Assistant App — Silver Prize & Graduation Project (2023.08)
Korea National University of Transportation
Video

Hanium: Automated Serving Robot using Object Detection & SLAM — Bronze Prize (2020.08)
Hanium ICT Mentoring Program, hosted by the Ministry of Science and ICT (MSIT), Korea
Video

Drift-adaptive Representation Preservation for Domain-Incremental Learning
Hyunmin Kong, **Gyutae Oh**, Jitae Shin
ACCV, 2026

Change Detection in Longitudinal Chest X-rays Is Readout-Limited, Not Task-Limited
Seungin Baek, **Gyutae Oh**, Jitae Shin
ACCV, 2026 · Oral

Routing Prompts Across Layers: A Global Memory Module for Prompt-Based Continual Learning
Gyutae Oh*, Jitae Shin
International Conference on Computer Research and Development (ICCRD), January 24, 2026

Multimodal Fusion Framework Using Contrastive Learning for Exposure Keratopathy
G Oh*, Y Won, D Lim, J Shin
International Workshop on Ophthalmic Medical Image Analysis (MICCAI 2025 Workshop), 2025
Paper **Code**

UNDER REVIEW

CVPR 2027
3 anonymous submissions under review

ICLR 2027
1 anonymous submission under review

JBHI 2026
1 anonymous submission under review

SELECTED PROJECTS

NRF Core Research: Adaptive State Inference & Real-Time Monitoring of Short-Term Deterioration
2026.09 – 2028.09
National Research Foundation of Korea (NRF)
Developing adaptive inference from partially available multimodal data and real-time monitoring of short-term deterioration.

Exposure Keratitis Diagnosis & Severity Prediction (Year 2)
2025.10 – 2026.10
Sungkyunkwan University · Kangbuk Samsung Hospital collaboration
Completed the second phase of research focused on model compression and performance improvements for practical use.

AI CHAMPION
2025.07 – 2025.09
Fundus-image-based multi-disease screening project with cross-lab and hospital collaboration; selected as a Top 100 team out of approximately 500 teams.

Campus Contest: Automated Serving Robot using Object Detection & SLAM — Grand Prize (President's Award) (2020.09)
Korea National University of Transportation
Video

SCHOLARSHIPS

SKKU BK21 Fellowship (Specialized Scholarship) (2024)
Sungkyunkwan University

University Innovation — Merit & Living Expense Scholarship (2024)
Sungkyunkwan University

Academic Excellence Scholarship (awarded 5 times, 2019–2023)
Korea National University of Transportation

PATENT

Device and Method for Diagnosing Exposure Keratopathy based on Multimodal Fusion Learning
Application No. 10-2025-0121342 · Filed Aug 28, 2025
Two-stage multimodal fusion learning framework for automatic grading of Exposure Keratopathy (EK). Trained with four ocular imaging modalities collected at a tertiary hospital (broadbeam, slit-beam, scattering, and blue-light): modality-wise features are aligned and fused during training, while inference enables modality-agnostic classification using only a single broadbeam image to implicitly leverage information from other modalities.

CERTIFICATION

Advanced Data Analytics Semi-Professional (ADsP)
Issued by Korea Data Agency (KDATA) · Sep 2025
ID: ADsP-046012145

PROFESSIONAL SERVICE

Reviewer (2026)
IEEE International Conference on Image Processing (ICIP)

Updated from portfolio content on October 16, 2026.